# KARTHIKEYAN KUMARAGURUPARAN

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## SUMMARY

Backend and distributed-systems engineer: a year-long software engineering internship building production multi-tenant services (Node.js, Temporal, AWS), now completing an MS in Computer Science at USC. Systems projects on failure behavior, all open source with green CI. Seeking new-grad backend/infrastructure SDE roles, available May 2027.

## EDUCATION

**University of Southern California** — **Aug 2025 – May 2027**
*Master of Science in Computer Science — GPA: 3.53/4.00* — *Los Angeles, CA*
*Relevant Coursework: Distributed Systems, Algorithms & Data Structures, Database Management*

**Kumaraguru College of Technology** — **Nov 2020 – May 2024**
*Bachelor of Engineering in Computer Science and Engineering — GPA: 9.23/10.0* — *Coimbatore, India*

## TECHNICAL SKILLS

**Languages:** Go, Python, C++, TypeScript, SQL, Bash
**Backend & APIs:** Node.js, Express.js, REST APIs, gRPC, Protocol Buffers, Temporal, Microservices
**Cloud & DevOps:** AWS (Lambda, API Gateway, IAM, KMS), Docker, Jenkins, GitHub Actions, Ansible, Nginx, Linux
**Databases & Storage:** MongoDB, RocksDB, SQLite
**Distributed Systems:** Quorum replication, write-ahead logging (WAL), crash recovery, idempotency, fault injection, linearizability checking

## PROFESSIONAL EXPERIENCE

**Clar Technologies — Client: Skuchain** — **Jul 2023 – Jul 2024**
*Software Engineer Intern · Node.js, Temporal, AWS, Docker, Ansible, MongoDB* — *Coimbatore, India*

- Built Node.js/Express microservices for a multi-tenant supply-chain-financing platform, running multi-step financing operations as durable Temporal workflows.
- Built serverless onboarding and key-management flows on AWS Lambda (API Gateway, IAM, KMS) serving 500+ users, and documented 8+ partner REST APIs.
- Cut environment provisioning from ~60 minutes to ~5 with Ansible; standardized Docker images and Jenkins delivery pipelines across a distributed team.
- Operated the platform's Nginx reverse proxy for 12 months without a recorded outage.

**Samsung PRISM — Samsung R&D Institute India** — **Mar 2023 – Sep 2023**
*Project Intern — raplscope: Linux System Energy Profiler · Go, Linux* — *Remote*

- Created raplscope, a zero-dependency Go CLI measuring whole-system energy and power from Linux RAPL counters via powercap sysfs; since rebuilt end to end and published at github.com/KarthikeyanK1206/raplscope.
- Engineered wraparound-safe accounting: wrap-corrected deltas, boundary snapshots so sub-interval commands measure correctly, and package-only summation to avoid double counting.
- Added /usr/bin/time-style command wrapping (inherited streams, signal forwarding, exit-code propagation) with JSON/CSV export and a bundled HTML energy dashboard.
- Made hardware-bound logic testable without root or RAPL hardware by injecting the sysfs root; 12 tests run against a fake counter tree.

## PROJECTS

**NimbusVault — Adaptive Distributed Object Store** — **C++17, gRPC, Protocol Buffers, RocksDB · Group Project**

- Built the metadata plane: a RocksDB-backed write-ahead log replicated from a fixed-leader coordinator with quorum commits — deliberately scoped below full Raft (no election or membership changes).
- Designed adaptive replication: access windows classify chunks hot/warm/cold with asymmetric hysteresis, scaling between 2 and 5 replicas — a one-third cold-storage reduction by construction vs. a static factor of 3.
- Built two-stage replica reconfiguration: a quorum-committed START keeps the old replica set readable while new copies are made and verified; COMMIT then publishes the stable set.
- Validated with 83/83 unit tests and 10/10 live-cluster integration scenarios in GitHub Actions CI; a Wing–Gong checker accepted 60/60 keys in each recorded 6,060-op concurrent history, including during reconfiguration (single-machine runs).

**RAEF — Fault-Tolerant Execution Runtime for LLM Agents** — **Python, SQLite, LangChain, OpenRouter · Group Project**

- Designed a log-before-send transaction layer: every tool call is journaled to SQLite before dispatch under a deterministic SHA-256-derived execution ID, so a restarted agent reuses committed results instead of re-firing writes.
- Built verification-driven recovery: an ambiguous write parks in a durable pending state until a pluggable verifier proves it committed (reuse) or provably absent (replay once), else hands off to an operator.
- Recorded a nine-phase crash-injection matrix asserting side effects from the target's own commit counter: every phase ends with exactly one mock-target commit, zero duplicates — 47 tests, CI green on Python 3.12/3.13.

## PUBLICATION

S. Barath Sanjay, B. Harish, K. Karthikeyan, et al., "IoT Based Smart IV Drip Stand," *IEEE Conference*, Oct. 2021.